Robust statistics

MATH 213

Set-up

- Open Statkey in a new browser tab.
  - Under the leftmost heading *Descriptive Statistics and Graphs,* click *One Quantitative Variable.*
  - Notice that a dotplot is automatically displayed for the data in the Mammal Longevity data set (noted on the top left of the screen.)

- Open the Codebook for the Mammal Longevity dataset here to see what information is included in this data set. You will notice that the code book lists two numerical variables: gestation and longevity. The variable "longevity" is the one whose histogram we are viewing.

Group 1

Our task today is to explore what it means for a statistic to be robust.

<u>Definition:</u> A statistic is called **robust** or **resistant** if extreme observations have little effect on their values.

- **Question:** There is one main outlier in terms of longevity. Which animal has this outlying longevity and what is its value? *(You can click "Show Data Table" to see the dataframe, and/or you can hover over dots on the dotplot to see more information.)*
- **Answer: Elephant, 40**

Here, we will vary the value of the longevity for the elephant in a couple different ways, and we will see what happens to the summary statistics as we make these changes. When you are done copying and pasting, as requested below, spend a minute comparing the tables of statistics - what changed and what didn't?

1.Copy and paste below the original table of summary statistics, before any changes are made.

2. In StatKey, click Edit Data, and change the entry for Elephant from 40 to 10. Copy and paste below the resulting table of summary statistics.

3. In StatKey, edit the data to change the entry for Elephant from 10 to 150. Copy and paste below the resulting table of summary statistics.

Robust (resistant) measures of center and spread

Reminder: A statistic is called **robust** or **resistant** if extreme observations have little effect on their values.

- Which measure of center (mean, median,) would you consider to be robust and which is not robust?

- Which measure of spread (standard deviation, the quantiles Q1 = the 0.25 quantile, and Q3 = the 0.75 quantile,) would you consider to be robust and which is not robust?

More tips for describing distributions

For purposes of describing a numerical distribution:

- Mean (for describing center) and Standard Deviation (for describing spread) “go together” and are good for describing symmetrical or close to symmetrical distributions

- Median (for describing center) and the quartiles (for describing spread) “go together” and are better for describing heavily skewed distributions.